

FINTECH 545 ASSIGNMENT 1 REPORT

1 Reading the Shape of a Sample

1(a)

n	= 1000
mean	= 0.001104
variance	= 0.000096
std dev	= 0.009811
skewness	= -0.669837
excess kurtosis	= 2.357614

Normal

Ruled out (needs skew ≈ 0 , kurtosis ≈ 0): ruled out by skewness — ours is -0.67 , not ≈ 0 .

Lognormal

Ruled out (needs skew > 0 , kurtosis > 0 , positive support): ruled out by skewness — the sign is wrong; lognormal is always positively skewed, but ours is -0.67 .

Student's t, symmetric NIG

Ruled out (needs skew ≈ 0 , kurtosis > 0): the kurtosis condition holds, but ruled out by skewness — these families are symmetric (skew $\equiv 0$), and our sample skew of -0.67 is roughly 8.7 standard errors from zero, too large to be sampling noise.

Normal inverse Gaussian, skewed t, generalized hyperbolic

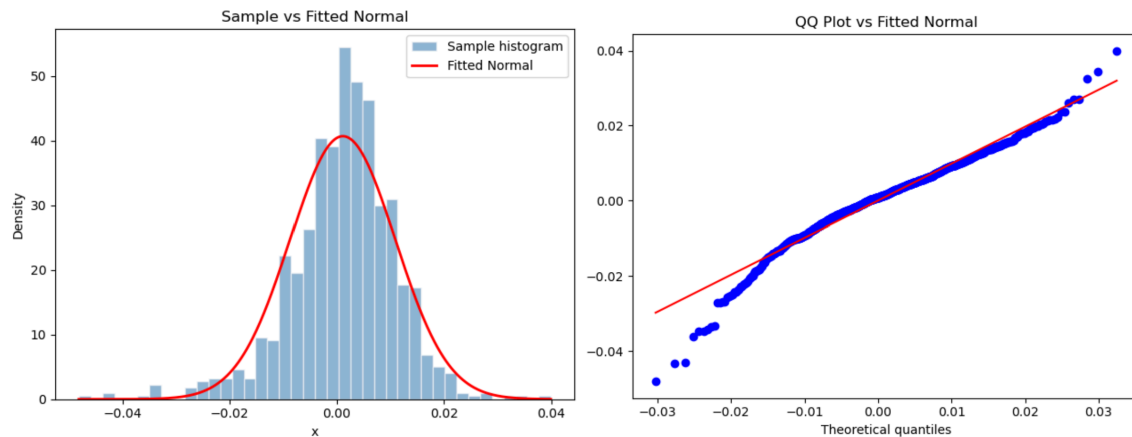
Not ruled out (needs skew not equals to 0, kurtosis > 0): both conditions are satisfied by the sample (skew = -0.67 not equals to 0, excess kurtosis = $2.36 > 0$), so this family remains plausible.

1(b)

Observed number = 26

Expected number = 10

1(c)



The error is concentrated in the tails, and it's asymmetric.

--- Left tail ---

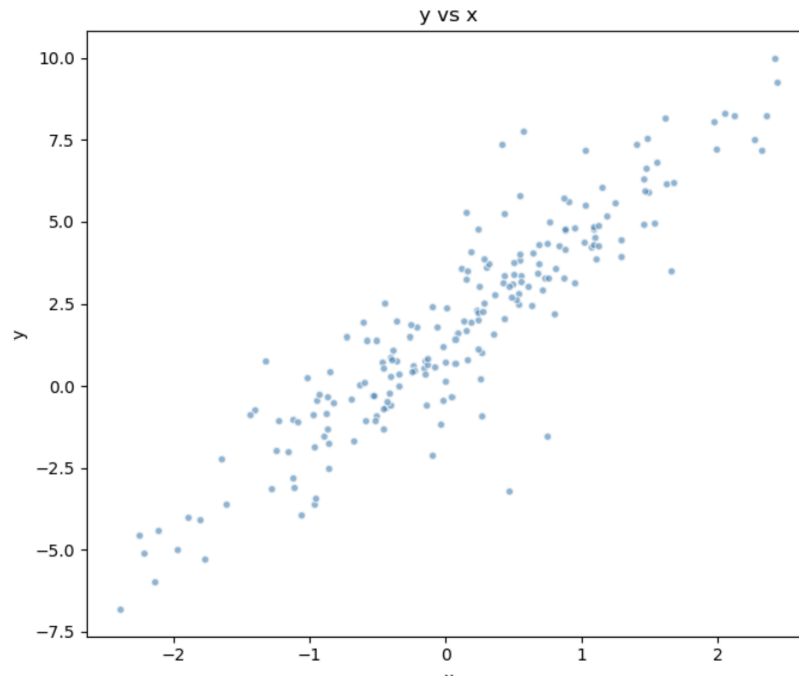
p=0.001	normal_q=	-0.02921	observed=9	expected=1.00
p=0.005	normal_q=	-0.02417	observed=18	expected=5.00
p=0.01	normal_q=	-0.02172	observed=26	expected=10.00
p=0.02	normal_q=	-0.01904	observed=34	expected=20.00
p=0.05	normal_q=	-0.01503	observed=50	expected=50.00

--- Right tail ---

p=0.95	normal_q=	0.01724	observed_above=34	expected=50.00
p=0.99	normal_q=	0.02393	observed_above=6	expected=10.00
p=0.995	normal_q=	0.02637	observed_above=5	expected=5.00
p=0.999	normal_q=	0.03142	observed_above=3	expected=1.00

- Left Tail: The Normal distribution greatly underestimates how often large negative values occur – at the extreme 0.1% level it's off by a difference of 8.
- Right Tail: the Normal roughly overestimates frequency near the 5% mark but underestimates the very extreme right tail (3 vs 1 at 0.1%)

2 A Regression Whose Errors Are Not Normal



2(a)

From the scatter plot, we can see that the relationship between x and y is approximately positively linear. The trend doesn't bend up or down at either end. However, there are some observations that are relatively far away from the main cloud/linear trend. So I predict the distribution to be Student's t, as it has heavier tails than a Normal distribution.

2(b)

Assumption 7: The error term is normally distributed.

To the estimated slope:

If the conditional mean is still correctly specified (the expected error term given x is 0), then even if the errors are not normal, beta will remain unbiased and consistent.

To the standard error of the slope:

OLS standard errors and the usual t-tests rely on the assumed error distribution, particularly in smaller samples. With heavy-tailed errors, extreme observations can make Normal-based hypothesis tests and confidence intervals less reliable.

2(c)

1. OLS

```
alpha (intercept) = 1.5607, SE = 0.0963  
beta (slope)      = 2.9908, SE = 0.0966  
sigma (resid std) = 1.3514
```

2. MLE Normal

```
alpha = 1.5607, beta = 2.9907, sigma = 1.3446  
log-likelihood = -343.0110  
AIC = 692.0221, AICc = 692.0462
```

3. MLE Student t

```
alpha = 1.5480, beta = 3.0192, scale = 0.9355, nu (df) = 3.6322  
log-likelihood = -328.5612  
AIC = 665.1224, AICc = 665.1626
```

The beta across all three models are close. The spread is 0.0285. This matches previous predictions, that the violation doesn't have an impact on the unbiasedness of slope. In the end, **Student-t is chosen as the best model for having the smaller AICc.**

2(d)

The Normal model is getting tails of the errors terms wrong. For normal distribution, extreme residuals are often rare. But Student-t's distribution allows for fatter tails. So AICc is rejecting the normal model for not being able to assign higher probability to extreme points.

2(e)

95.0% quantile:

Normal: 2.2117

Student-t: 2.0538

99.5% quantile:

Normal: 3.4636

Student-t: 4.6201

This occurs because the Student-t distribution has heavier tails, which become increasingly important at extreme quantiles.

I would use the Student t' model to ensure the bank has enough capital to counter worst-case risk scenarios. Being more conservative and getting a better estimate of extreme events are better for dealing with tail risks.

3 Pearson Against Spearman

3(a)

I think the pair x_1 and x_2 will disagree the most. Because the graph shows a perfectly non-decreasing pattern. Since Spearman accounts for monotonicity, the Spearman coefficient is likely 1, while Pearson will be far from 1 because the relationship is not a straight line.

On the other hand, x_3 and x_4 will likely agree on the most. Because there seems to be no correlation between x_3 and x_4 at all – we also can't tell if it's increasing or decreasing. So both Pearson and Spearman will be close to 0.

3(b)

```
---Pearson Correlation Matrix---
      x1      x2      x3      x4
x1  1.000000  0.799092  0.719467 -0.004680
x2  0.799092  1.000000  0.587614 -0.020923
x3  0.719467  0.587614  1.000000 -0.008605
x4 -0.004680 -0.020923 -0.008605  1.000000
---Spearman Correlation Matrix---
      x1      x2      x3      x4
x1  1.000000  0.971930  0.683910  0.000189
x2  0.971930  1.000000  0.667422 -0.001131
x3  0.683910  0.667422  1.000000 -0.002422
x4  0.000189 -0.001131 -0.002422  1.000000
```

The Pair with the largest gap is (x1, x 4)

3(c)

Spearman's correlation of 0.972 is the more honest description because the relationship is strongly monotonic but nonlinear; Pearson's 0.799 answers the narrower question of how strongly the two variables are linearly related.

4 Conditional Distributions

4(a)

X1 is the first block, x2 is the second block.

Answer as below: The conditional variance of x2 given x1 is around 1.4867. After observing x1, the uncertainty about x2 falls to 36.2% of its original level, corresponding to a 63.8% reduction in variance.

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{pmatrix}$$

$$\text{Var}(X_2|X_1) = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12} \\ \approx 1.4867$$

$$\frac{\text{Var}(X_2|X_1)}{\text{Var}(X_2)} = \frac{\Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}}{\Sigma_{22}} \\ \approx 0.3627$$

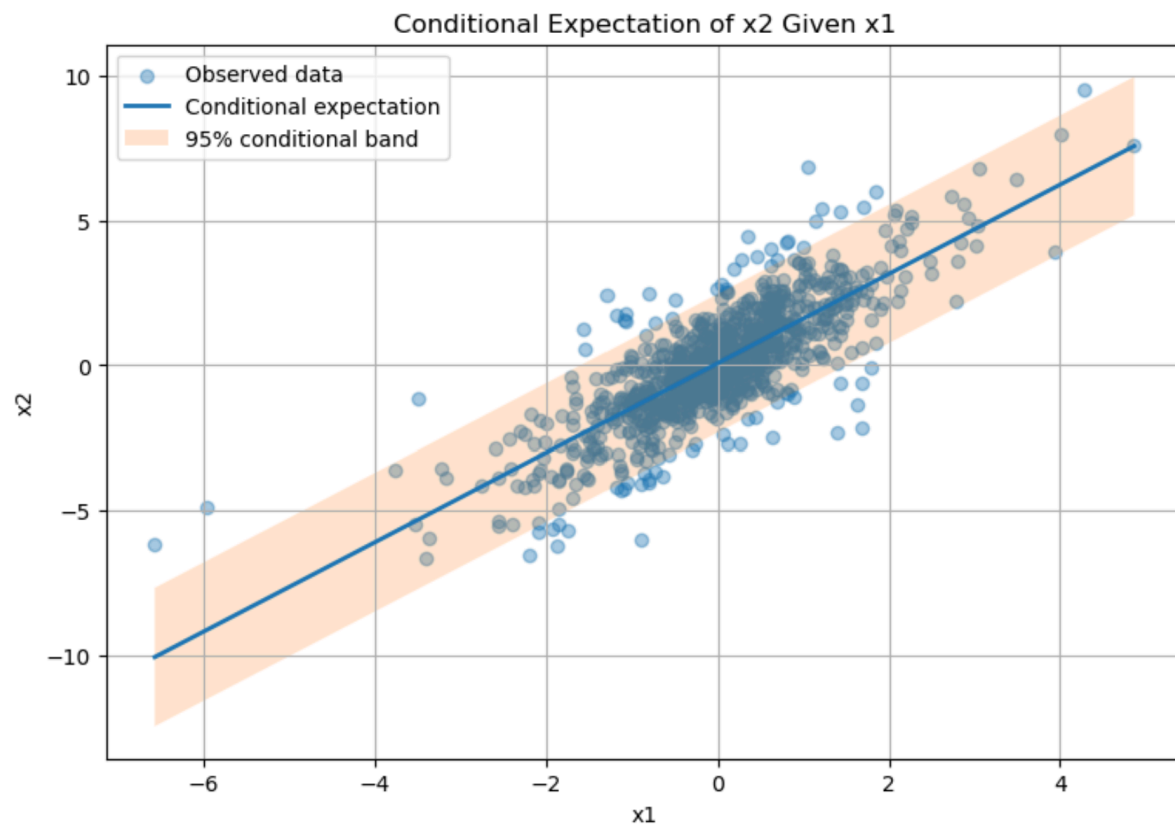
4(b)

Under the multivariate normal, the factor doesn't depend on the value of x_1 observed. The key is the conditional covariance formula in the picture above – $\text{Var}(x_2|x_1)$. The conditional variance depends only on the blocks of Σ , not on x_1 .

4(c)

$$\bar{\mu} = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (X_1 - \mu_1)$$

$\Sigma_{21} \Sigma_{11}^{-1}$ is the matrix of coefficients from regressing X_2 on X_1 .



Coefficient on x1: 1.5428133384439877
Conditional variance: 1.4867456782952875

4(d)

Fraction inside: 0.935

4(e)

Coverage by distance of x_1 from its mean:

Within 1 SD: 0.9565 (95.65%), $n = 759$

Between 1 and 2 SD: 0.8579 (85.79%), $n = 190$

Beyond 2 SD: 0.9020 (90.20%), $n = 51$

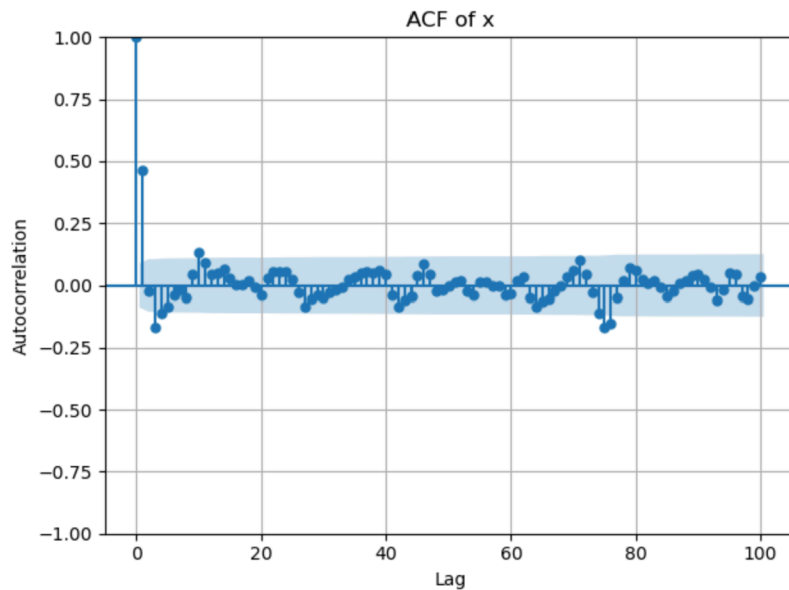
4(f)

The fact that the coverage is not constant across the three x_1 buckets shows that the variance of x_2 changes depending on x_1 . The assumption that $\text{Var}(X_2|X_1) = \text{constant}$ in the constant-width 95% band in part (c) has failed. However, the conditional mean from part (c), and its regression slope, still survives because the mean relationship can remain linear. This is because the coefficient Σ_{21}/Σ_{11} is still the slope of the linear regression of x_2 on x_1 .

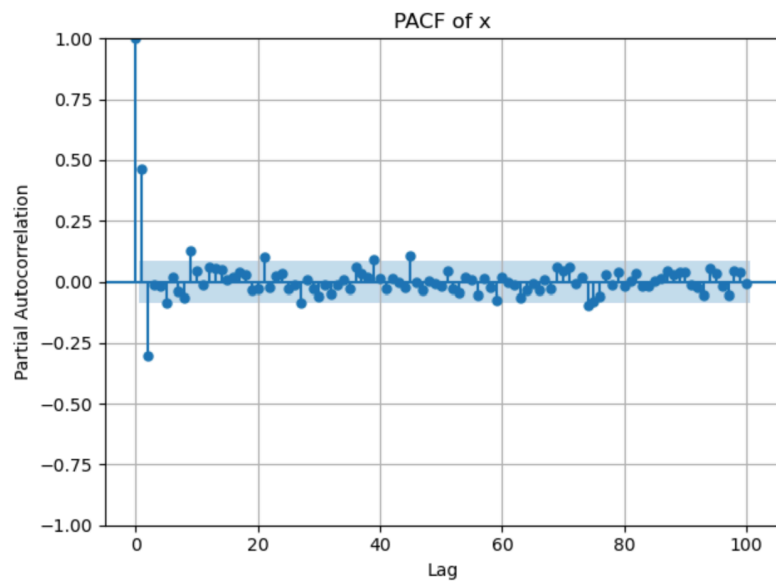
What fails is the assumption that the conditional variance is constant, so the constant-width confidence band from part (c) is no longer appropriate.

5 Identifying an AR or MA Order

5(a)



ACF: large significant spike at lag 1, then it largely drops into the significance band → **cuts off**



PACF: after lag 1, it generally stays around zero with some small fluctuations → **decays**

Therefore, I believe the pattern is most consistent with **MA(1)**.

5(b)

I used the rule that an ACF/PACF spike is considered statistically significant if it falls outside the approximate 95% confidence band.

5(c)

```
AR(1) : 1416.3653
AR(2) : 1368.3391
AR(3) : 1368.4262
MA(1) : 1389.0843
MA(2) : 1381.2316
MA(3) : 1374.1165
```

5(d)

According to the result from the data, AR(2) has the lowest AICc, which makes it different from my prediction (MA(1)).

5(e)

The AR(3) model adds a third lag coefficient, but this coefficient is small, so the additional lag provides only a small improvement in fit. AIC charges 2 per parameter as a penalty for model overcomplexity. The small improvement from AR(3) is not enough to overcome this penalty, so AICc selects AR(2).

R^2 does not penalize additional parameters, so adding the third lag would cause R^2 to increase or remain unchanged. Therefore, R^2 would not necessarily reject the larger AR(3) model.